

How to Stop Deepfake Identity Impersonation Attacks

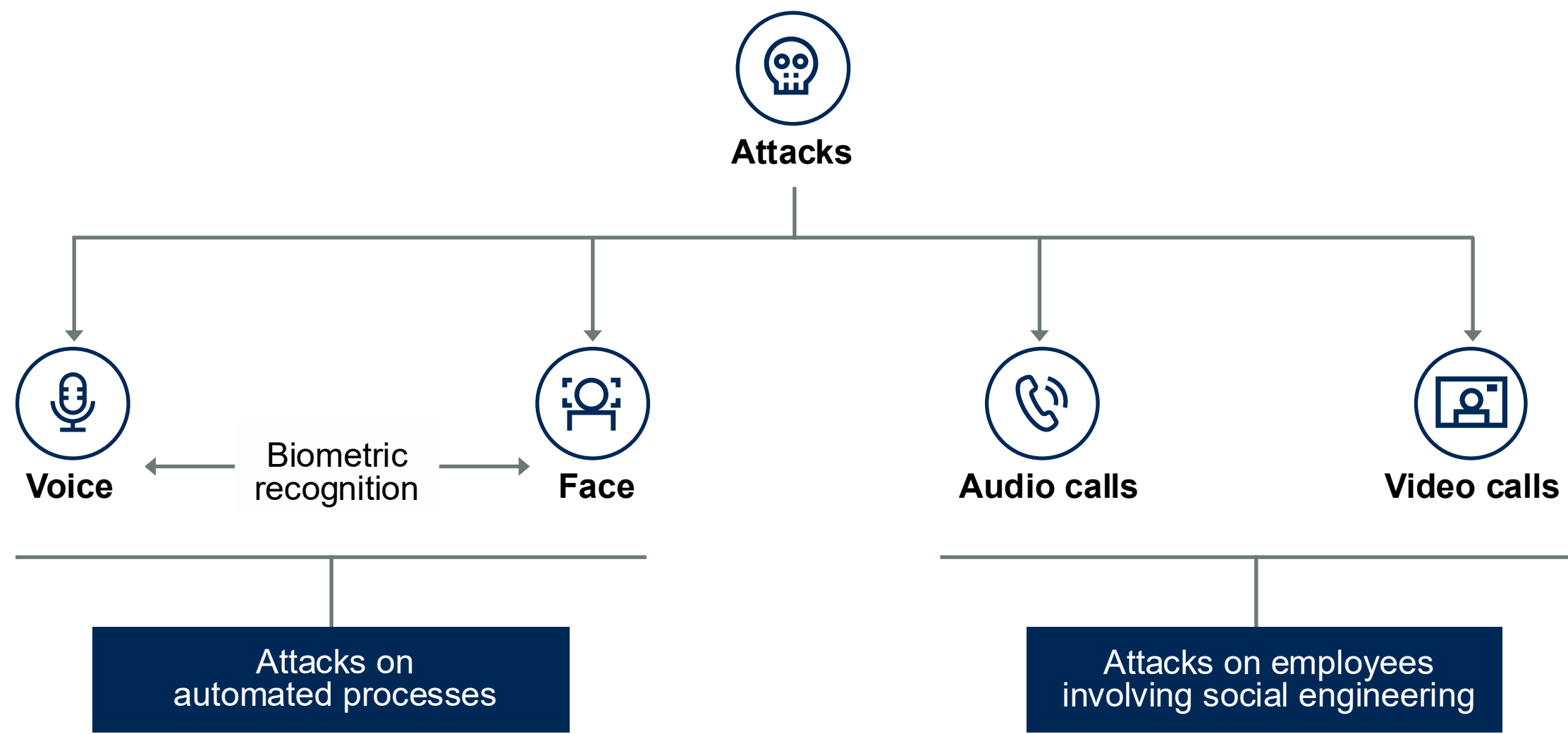
Akif Khan

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Deepfake Identity Impersonation



Source: Gartner

How Common Are Such Deepfake Attacks?

	We did not experience this	We had at least one minor incident	We had >5 minor incidents	We had at least one major incident	We had >5 major incidents
Deepfake audio used against automated voice biometrics	67%	19%	11%	2%	0%
Deepfake video used against automated face biometrics or identity verification	69%	19%	9%	2%	1%
Social engineering with deepfake during audio call with an employee	56%	28%	10%	5%	0%
Social engineering with deepfake during video call with an employee	63%	21%	10%	5%	1%

n = 302

Major incident = resulting in loss of IP, significant financial loss or business interruption

The Attack Surfaces



**Voice
biometrics**



**Face
biometrics**



**Social engineering
of employees**

The Attack Surfaces



**Voice
biometrics**



Face
biometrics



Social engineering
of employees

Voice Biometrics Use Cases

- Contact center (typically financial services).
- Some IT help desks.
- Automated authentication and reducing agent handling times.



Attacks are not new — e.g., voice replay attacks

Creating Deepfake Voice Today

- Voice cloning services easily available.
- Sample times vary from <1 min to >30 mins, depending on desired quality.
- Text-to-speech features.
- Few protections against cloning another person's voice.

The image displays three overlapping promotional cards for voice cloning services, each with a torn-edge border. The top card is white and features a microphone icon and the text 'VOICE CLONING' in a small, bold, sans-serif font. Below this, the headline 'Create a replica of your voice that sounds just like you' is written in a large, bold, black sans-serif font. A sub-headline at the bottom of the card reads 'Automate video voiceovers, ad reads, podcasts, and more, in your own voice' in a smaller, regular sans-serif font. The middle card is light gray and has the headline 'AI Voice Cloning that sounds human' in a large, bold, dark gray sans-serif font. Below the headline, a paragraph of text states: 'Over 1,000,000 users create high their voice by deploying the most voice cloning model. Use your own unparalleled control over synthetic capture human emotion in stunning'. The bottom card is also light gray and features the headline 'AI VOICE CLONING' in a large, bold, dark gray sans-serif font. Below the headline, a paragraph of text reads: 'Create lifelike AI voiceovers in your voice—no recording every time. Scale your videos, podcasts, and presentations while keeping your personal touch'. At the bottom of this card is a prominent blue rounded rectangular button with the white text 'Clone your voice →'.

VOICE CLONING

Create a replica of your voice that sounds just like you

Automate video voiceovers, ad reads, podcasts, and more, in your own voice

AI Voice Cloning that sounds human

Over 1,000,000 users create high their voice by deploying the most voice cloning model. Use your own unparalleled control over synthetic capture human emotion in stunning

AI VOICE CLONING

Create lifelike AI voiceovers in your voice—no recording every time. Scale your videos, podcasts, and presentations while keeping your personal touch

Clone your voice →

Source: [ElevenLabs](#); [Resemble AI](#); [VEED](#)

Deepfake Voice Detection



Analysis of perceptual features: Signal processing analysis of amplitude range, pitch deviation, cadence and others.



Spectral feature-based detection: ML-based analysis of features such as spectral centroid, contrast and flux.



Deep neural network: Training on huge datasets to allow the DNNs to learn and identity artefacts used for inference/classification.



Machine vision: Translation of audio spectrograms into images which are then classified using vision-based ML or DNN models.

Generalization remains a problem, performance drops when tested against tools not used in training.

Accuracy Is Hard to Assess

There are currently no standards for assessing efficacy of deepfake voice detection.

99% accuracy when detection models have been trained on samples from the voice cloning tool

85 to 90% accuracy for “zero-day attack” using a voice cloning tool that is new to the detection models

Vendor claims made to Gartner that Gartner cannot validate

Most organizations make selection based on ease of integration with existing voice infrastructure (e.g., Genesys, Avaya, Five9 etc.).

Source: Claims from five vendors made during briefings in Q4 2024

Protect Voice Biometrics With Layers

Voice authentication

Deepfake voice detection

Additional authentication factors

Caller ID spoofing detection

SIM swap detection

Phone/identity correlation services

Combine the (often probabilistic) signals from all of these to reach an overall decision about the integrity of the call.



Deepfake voices are a threat to automated voice authentication. Check if your authentication vendors have native deepfake voice detection capability.

The Attack Surfaces



Voice
biometrics



Face
biometrics



Social engineering
of employees

Face Biometrics Use Cases

Out of scope: OS-native biometrics such as Apple FaceID or Windows Hello for Business — security leaders have little control over the technology involved

Most common use case is as part of online identity verification (IDV):



Picture of identity
document assessed
for authenticity



Selfie assessed for liveness
then biometrically compared
to identity document

Where Is IDV Used in Your Business Today?

Customer/citizen

Regulated (KYC) onboarding
Government services
Healthcare
Marketplaces
Gig economy

Deepfake impact: **medium**

Stolen funds/PII (one account at a time),
fraud, fines and unhappy customer.

Workforce

Account recovery
Workforce onboarding
Post IDV selfie
-based authentication

Deepfake impact: **high**

Single deepfake attack could lead to
ransomware, stolen IP and large funds theft.

Creating Deepfake Face Today

- Face swapping services readily available, including real-time ones.
- Services to create AI avatars based on uploaded images and video.
- **>7000** “deepfake” projects in GitHub including offensive toolkits.



Source: [Deepfake Forum & Creator Community](#); [HeyGen](#); [GitHub](#)

Using the Deepfake — Presentation Attacks



Deepfake
possible

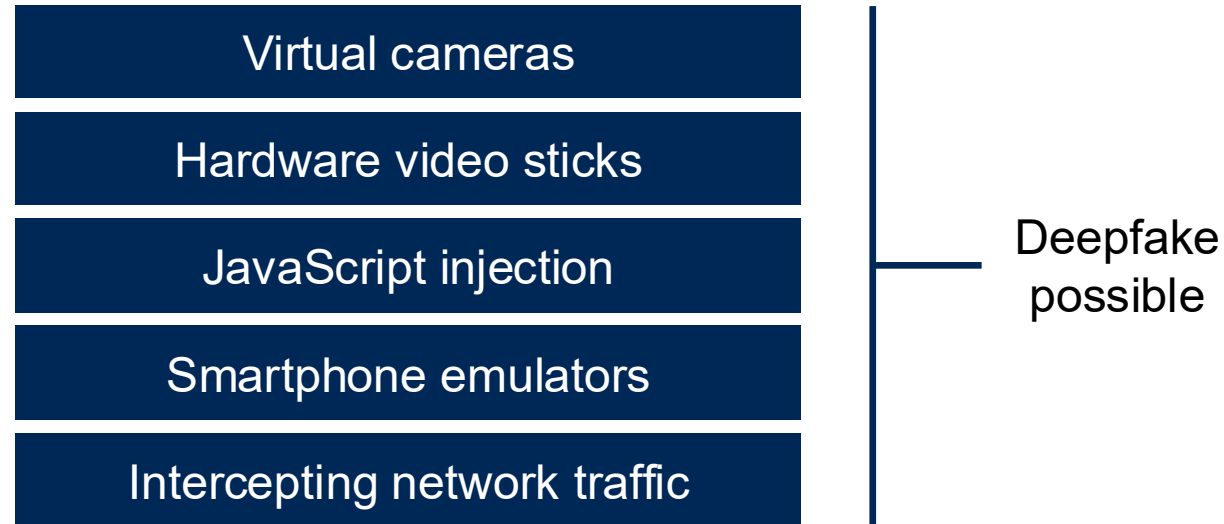
Photo or photocopy
Printout
Tampered document
Counterfeit document
Image displayed on screen

Photo
Paper mask
Hyper-realistic mask
Image displayed on screen
Video displayed on screen

Deepfake
possible

In presentation attacks, the fraudulent artifact is presented to the physical sensor (camera).

Using the Deepfake — Injection Attacks



In injection attacks, digital content is introduced into the IDV process by-passing the physical sensor (camera).

Deepfake Face (Image/Video) Detection



Analysis of image features : Facial landmark alignment and mobility, blink analysis, inconsistent or repeated backgrounds/clothing/glare/shadows.



Neural networks : CNNs to detect pixel level anomalies; RNNs to detect temporal inconsistencies between frames.



GAN/diffusion fingerprint detection : Identifies noise patterns in frequency and spatial domains that result from use of GAN/diffusion image generation models.

Similar to audio, generalization remains a problem,
performance drops when tested against tools not used in training.

Analysis of image features provides some mitigation against
this as it is independent of the model used to create the image.

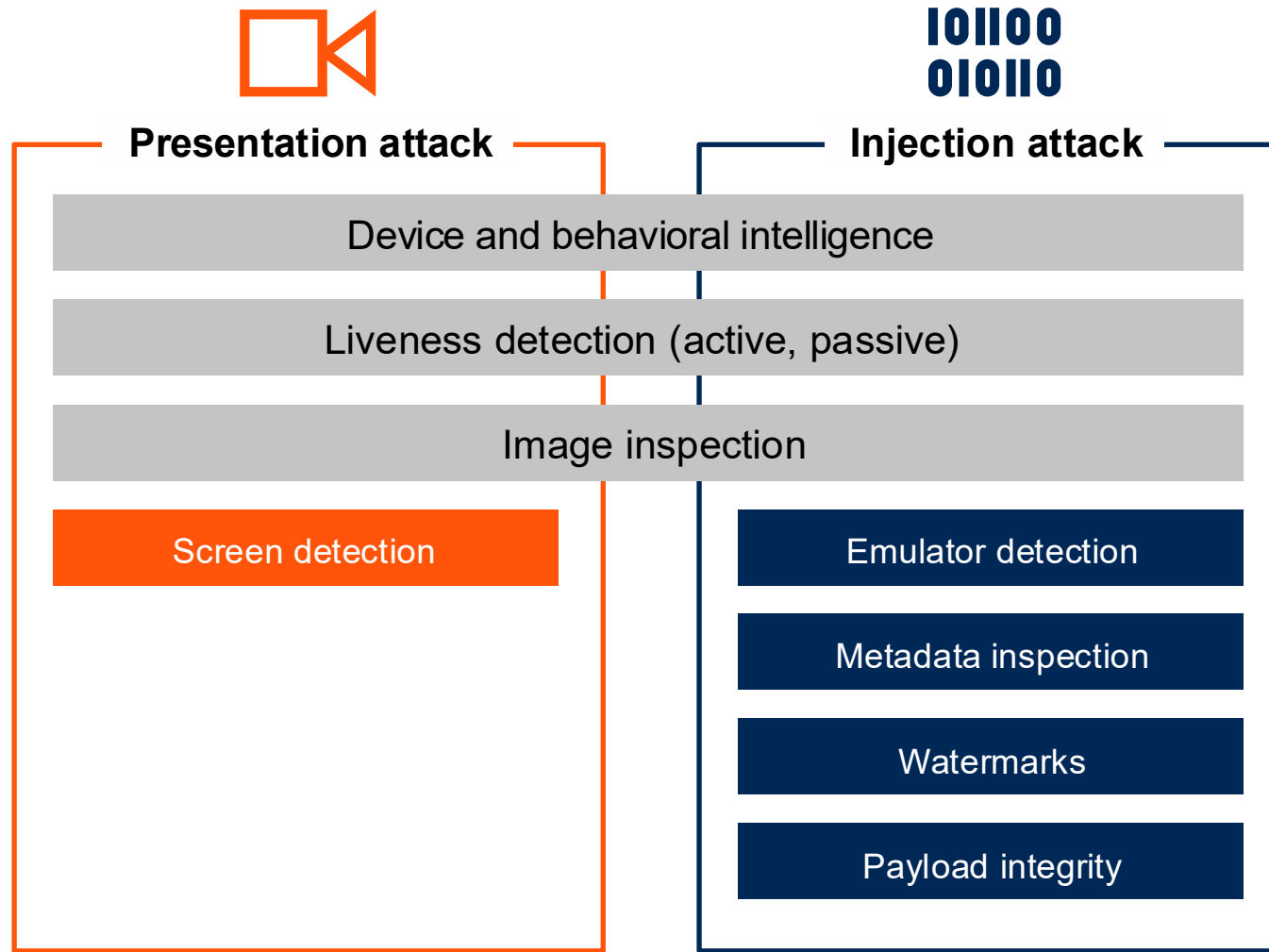
Layered Defensive Capabilities

Capability	Description
Device and behavioural intelligence	Assessing device telemetry and velocity of use; comparing interactions with the user interface with statistical norms.
Liveness detection	Active or passive techniques to assess genuine human presence when selfie is taken.
Image inspection	Applying computer vision models to the image to look for signs of GenAI use, also looking for repeated features such as backgrounds or light patterns.
Screen detection	Detecting signs that camera is being pointed at a screen, such as glare, reflections or pixelation.

Layered Defensive Capabilities

Capability	Description
Emulator detection	Detection of virtual cameras, virtual machines and virtual mobile devices being run on larger machines.
Metadata inspection	Examination of image time stamps, comparing image resolution with that expected from camera of given device.
Watermarks	Addition of visually obfuscated watermarks to image when taken by vendor's SDK, the absence of which would infer that the SDK was not controlling the image capture.
Payload integrity	Strong cryptographic protection of the image payload being sent from vendor's SDK to its SaaS environment

Seek Vendors Who Bring It All Together



Accuracy Is Hard to Assess

ISO 30107-3:2023

Describes how to test a system for PAD. Most vendors in the market today have been tested accordingly.

CEN/TS 18099:2025

Just introduced for IAD, now waiting for biometric labs to offer certification testing.

Favour vendors that examine the broadest set of risk signals, maximizing the chance that they can detect identity impersonation, even if they fail to detect the deepfake itself.



Preventing deepfake identity impersonation attacks is not just about being able to detect the deepfake — all risk signals are useful.

The Attack Surfaces



Voice
biometrics



Face
biometrics



**Social engineering
of employees**

A Growing Problem

UK engineering firm Arup falls victim to £20m deepfake scam

Hong Kong employee was duped into sending cash to criminals by AI-generated video call

'I Need to Identify You': How One Question Saved Ferrari From a Deepfake Scam

- Benedetto Vigna was impersonated on a call using AI software
- Large companies are being increasingly targeted with deepfakes

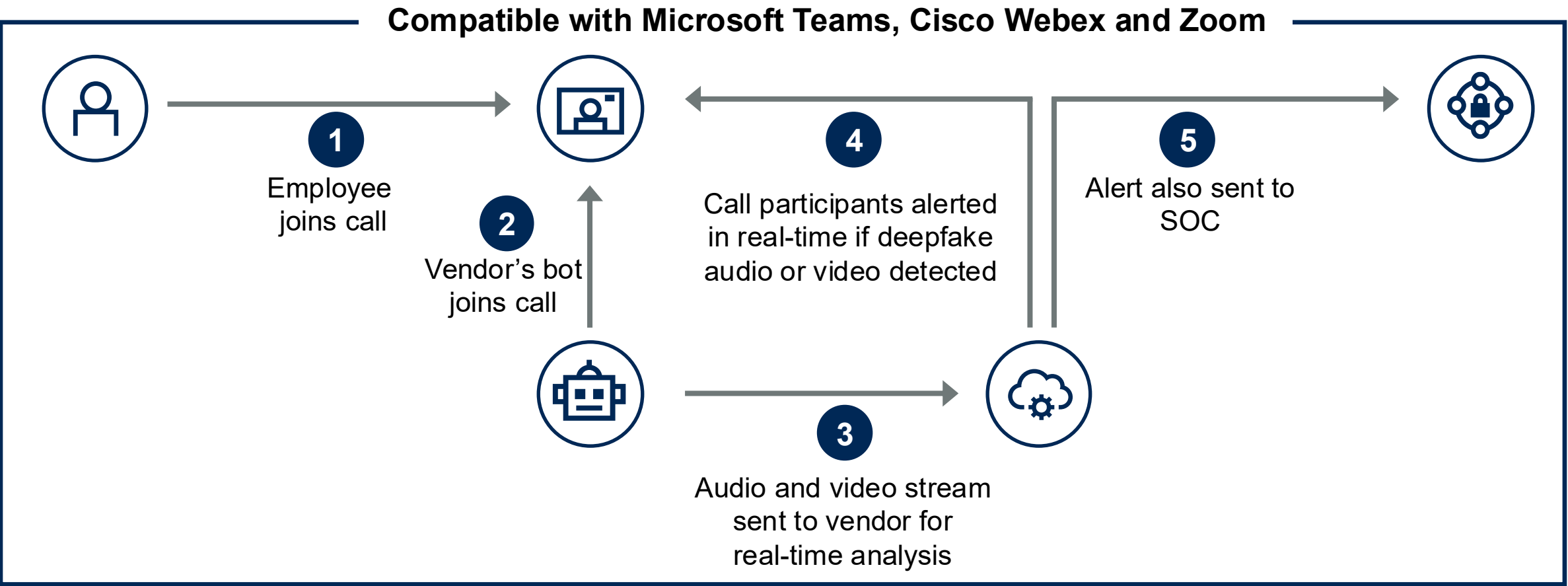
CEO of WPP, world's biggest advertising agency, falls victim to elaborate deepfake scam

Deepfake scams escalate, hitting 53% of businesses

Eighty-five percent of corporate finance professionals now view such scams as an “existential” threat, a Medius study found.

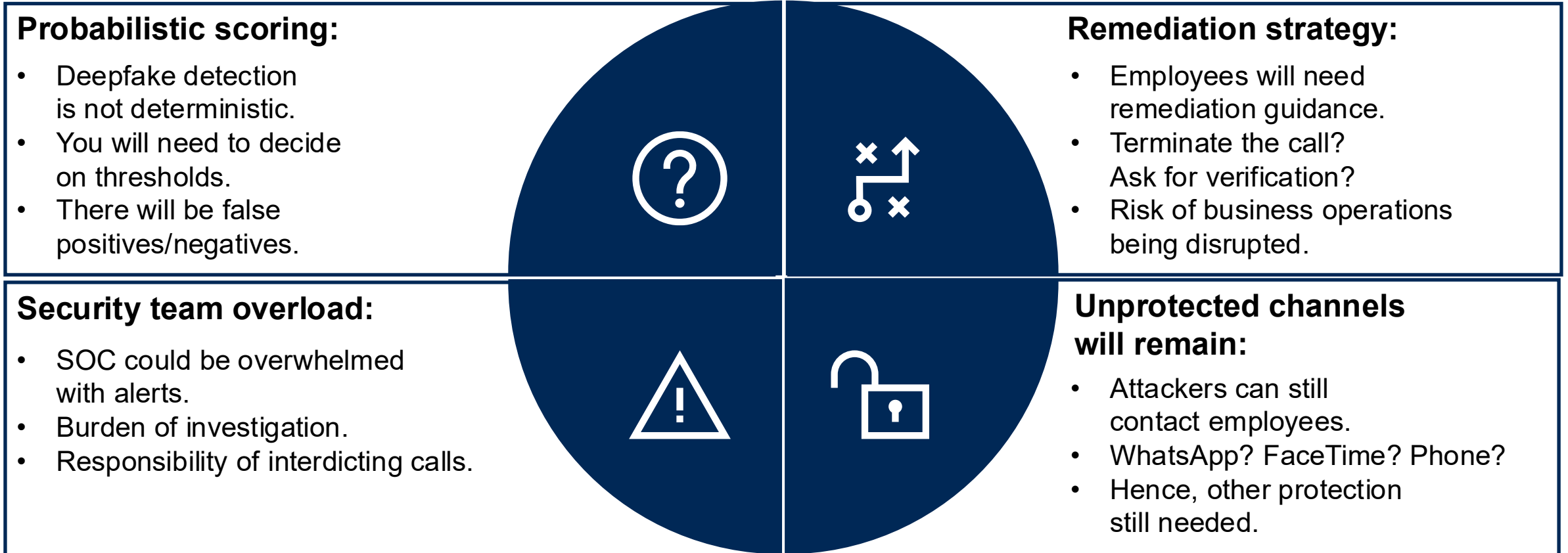
Source: [UK engineering firm Arup falls victim to £20m deepfake scam](#), The Guardian; [CEO of WPP, World's Biggest Advertising Agency, Falls Victim to Elaborate Deepfake Scam](#), New York Post; ['I Need to Identify You': How One Question Saved Ferrari From a Deepfake Scam](#), Bloomberg; [Deepfake Scams Escalate, Hitting 53% of Businesses](#), CFO Dive

New Vendors Introducing Technical Solutions



Example start-up vendors:
DeepTrust; GetReal; Netarx; Reality Defender; Resemble AI

Challenges With This Technical Approach



Focus on Fundamental Layers of Defense



Employee behaviour

- Educate and train employees about the nature of the threat
- Explore using security education vendors that focus on deepfake threats
- Encourage low-tech approaches such as asking personal questions



Business process hardening

- Audit business processes to find those most vulnerable to this threat
- Add systemic authorization steps to high-risk processes
- Reduce the chances that a single phone call can lead to catastrophe



Authentication & verification flows

- Implement processes to allow employees to authenticate themselves
- Use phishing-resistant MFA like FIDO2 or X.509 to authorize risky events
- Implement automated IDV for account recovery workflows



Deepfakes and social engineering are a bad combination — make your people and processes more resilient but stay aware of nascent technical solutions.

A Final Word ...



Deepfake detection
is an **emerging**
art/science.



The attack vector
is evolving at an
incredibly rapid pace.



Mitigate risk by
assuming that deepfake
detection will **fail**.

No matter how well it performs at a given point in time, deepfake detection should not be relied upon entirely and should always be combined with other risk signals.

Action Plan for Security Leaders

Monday morning

- **Confirm** where in your business you use automated biometric voice authentication and biometric identity verification.
- **Check** if there has been any informal reporting of social engineering attacks involving deepfake audio/video.

Next 90 days

- **Review** the capabilities of your biometric vendors to check for deepfake detection plus other risk signals. **Start** new procurement if needed.
- **Engage** stakeholders to run an organization wide audit of business processes most vulnerable to deepfake social engineering attacks.

Next 12 months

- **Harden** vulnerable business processes to reduce the impact of a malicious call.
- **Implement** robust MFA to authorize high-risk activities and IDV for account recovery.
- **Monitor** the emerging market for deepfake detection in video calls.

Recommended Gartner Research

To learn more about access to Gartner research, expert analyst insight, and peer communities, contact your Gartner representative or click on “Become A Client” on gartner.com to speak with one of our specialists.

🔍 **[How to Mitigate Deepfake Identity Impersonation Attacks](#)**

Akif Khan

🔍 **[Magic Quadrant for Identity Verification](#)**

Akif Khan, James Hoover and Nayara Sangiorgio

🔍 **[Quick Answer: How to Secure User Credential Management and Account Recovery](#)**

James Hoover, Ant Allan and Akif Khan